Jom (Pornthep) Preechayasomboon

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Education

Expected 2021	Ph.D. Mechanical Engineering University of Washington GPA 3.90/4.00
2016 – 2018	M.S. Mechanical Engineering University of Washington GPA 3.90/4.00
2011 – 2015	B.Eng. Mechanical Engineering Chulalongkorn University (Thailand)

Skills

› Development	C#, C++, Unity3D, Python, Julia, ROS, MATLAB, bash
› Technical	Hardware development, Hardware-software integration, Rapid prototyping, 5-axis CNC machining, Microcontrollers and electronics, Molding and casting
› Engineering	SolidWorks, Catia V5, Siemens NX ANSYS (Structural & Fluent), Altium Designer, CUDA

Interests

Soft Robotics, Haptics, Virtual Reality, Machine Learning, Additive Manufacturing, Game Development

Work Experience

5/2017 – Present **Research Assistant**
Rombolabs (Dr. Eric Rombokas), University of Washington

Multi-modal sensors and digital fabrication in soft robotics
- My PhD research revolves around the creation of novel ways to fabricate and reason with sensors for soft robots.

Identifying the metrics of the Rubber Hand Illusion with a novel perceptual science experiment
- I integrated multiple systems (PHANToM haptic robot, Oculus Rift, LEAP hand tracking, Shimmer GSR+ body sensors) using ROS and Unity3D as a backend to simulate rich visuotactile sensations in a knife-game setting, performed human subject studies and extensive data analysis on human kinematics and automatic nervous responses

4/2019 – 9/2019 **Research Intern (Haptics)**
Oculus, Facebook Reality Labs (Redmond, WA)

Working with Ali Israr, I worked on the hardware and software development of a novel compact, leadscrew-based haptic actuator, “Chasm”. I designed and developed almost all aspects of the project, including the firmware and software for running haptic demos with the actuator. Chasm is to be presented at CHI2020 in April of 2020.

8/2015 – 8/2016 **Research Engineer**
Biomechanical Design and Manufacturing Laboratory, Chulalongkorn University (Thailand)

Hip implant manufacturing
- Led the development of a piezoelectric and flexure driven tooling system for micron-tolerance machining of femoral head prosthesis

3/2014 **Manufacturing Assistant**
Biomechanical Design and Manufacturing Laboratory, Chulalongkorn University (Thailand)
- Aided in running 5-axis CNC machinery to produce BDML’s polycentric prosthetic knees

Publications

2020 Negshell Casting: Strong, Lightweight and Complex Soft Robot Fabrication Using SLA Printing and Traditional Casting
(In preparation)

2020 Chasm: A Screw Based Expressive Compact Haptic Actuator
(Presenting at CHI2020, Hawaii, USA – 24.3% Acceptance Rate)
Manuscript and video available upon request

2019 ConTact Sensors: A Tactile Sensor Readily Integrable into Soft Robots
(Presented at 2019 IEEE International Conference on Soft Robotics, Seoul, Korea)
IEEE Xplore, Full Paper, Video

Notable Projects & Extracurricular Activities

2019 ovrseer – A framework for VR psychophysics experiments in Unity3D

- ovrseer came out of the necessity of rapid prototyping psychophysics experiments. ovrseer's backbone is a finite-state-machine where experimenters can simply drag and rearrange "phases" of an experiment to iterate between one experiment to the next.

2018 DroidPool – An exploration of creating game artificial intelligence with Unity3D's ml-agents

- Developed a small game to train agents in games to behave similarly to humans in a multiplayer setting using deep reinforcement learning

2016 Scapl – CUDA based voxel editing software

- Developed software that can edit massive monochrome voxel data used for 3D printing intricate microstructures on Stratasys Objet printers. Slicing is done directly in SolidWorks and is then modified using an OpenGL interface with CUDA as a parallel computing voxel editing and rendering back-end.

2015 Development of an Electric Prosthetic Hand with an Anti-Slip Tactile System – Senior Project

- Developed a novel tactile sensor capable of slip detection based on a magnetic displacement sensor embedded in a hyper-elastic elastomer
- Designed and CNC machined a robotic prosthetic hand based on a brushless DC motor and a novel sliced bevel gear design
- Programmed a system for performing repeated grasping tasks using a robotic arm and the robotic prosthetic hand by integrating Processing, Arduino and Maxon motor controllers.

Awards & Achievements

2018 Honorable Mention for the Soft Robotics Toolkit Competition in Contribution to Soft Robotics Research

- ConTact Sensor is featured as one of the Sensor modules in the Soft Robotics Toolkit softroboticstoolkit.com/contact-sensor

2015 Best Senior Project Award

- Won the annual best senior project award amongst 43 projects in the Mechanical Engineering department

Presentations

2019 ConTact Sensor was presented at RoboSoft 2019, Seoul, Korea as a podium presentation

2018 Motor Behavior Changes in Response to the Rubber Hand Illusion was presented at the Neural Computation and Engineering Connection, University of Washington

Personal Projects

2018 ROSCsereal – A Unity3D plugin for communicating with the Robot Operating System
github.com/prnthp/ros-csereal-plugin